

Problem Statement

Portfolio Governance Bot — 05-portfolio-governance-bot

The Problem / Opportunity

Portfolio and program managers spend a disproportionate amount of time manually re-reading a tracker to answer the same handful of questions over and over: what's at risk, what's slipped, what needs governance attention, what's blocked and why. That information already exists in a structured sheet; the bottleneck is retrieval and synthesis, not data collection.

This project sits inside a broader assignment: build a Portfolio-governance system: a pipeline that pulls information from multiple sources, processes each source through a specialized AI agent, and synthesizes everything into a single, actionable workflow delivered to a communication channel of choice. The orchestration mirrors the cross-functional coordination that happens in any real program: decomposing an ambiguous problem into clear ownership, routing work to the right specialist, aggregating outputs, and ensuring the whole thing is reliable even when one piece fails.

Who's Affected

Portfolio managers and senior program stakeholders who need fast, accurate, governance-aware answers about program status; without personally re-reading a tracker every time a question comes up, and without waiting on whoever last updated the sheet to be available to answer.

Workflow Elements

- Webhook triggers: reliably receiving an event from an external system and correctly acknowledging it, including protocol-specific handshakes like Slack's one-time URL verification.
- Input filtering / normalization: telling a real request apart from noise, and reshaping different-shaped inputs into one common contract.
- Live data fetching: pulling current state from a system of record rather than working from stale or hardcoded data.
- A single AI agent reasoning over that data: constrained to the data it's given, producing a consistent, audience-appropriate answer.
- End-to-end error handling: a failure anywhere in the chain surfacing as a visible alert, not a silent drop.

Each of those is a prerequisite to build the workflow.

What This Project Solves

- Answers natural-language portfolio-status questions: risk, timeline slippage, governance level, open defects; from a live Google Sheet.
- Accepts questions from two structurally different entry points (a Slack Events webhook and a standalone HTML form) through one shared normalization layer.
- Routes a finished answer to the correct destination (a Slack thread, an HTTP response, or both) as a distinct step from generating that answer.
- Surfaces any pipeline failure to a dedicated Slack alert channel via a single, workflow-wide error handler.